TRIVIAL H^0 ON FIBRES SPLITS OFF THE DERIVED PUSHFORWARD OF THE STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-flat-proper-perfect-direct-image-general is same-chapter NOT carded (cross-line-claimed label) and is declared external.

Allowed standard prerequisites (editorial).

- Perfect pushforward with base-change compatibility. lemma-flat-proper-perfect-direct-image-general (same chapter, NOT carded, declared external). E = Rf_*O_X is perfect, base-change compatible, with tor amplitude in [0, infinity).
- Cutting a perfect complex in two at the H^0 level. More on Algebra, lemma better-cut-complex-in-two (cross-chapter, named). Locally on S, E = H^0(E) + tau_{>= 1}E with tau perfect of tor amplitude [1, infinity).
- Perfect flat modules are finite locally free. More on Algebra, lemma perfect; Algebra, lemma finite-projective (cross-chapter, named). H^0(E) is finite locally free.
- Flat cokernel criterion. Algebra, lemma cokernel-flat (cross-chapter, named). O_S -> H^0(E) is checked after tensoring with residue fields.

Source reference. Original reference 00O0: focused-seeds/geometry/S9-perf2-proper-flat-h0/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00NX: focused-seeds/geometry/S9-perf2-proper-flat-h0/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0B91: sources/stacks-perfect/pinned-original-source.tex, source lines 7742–7742. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A1U: focused-seeds/geometry/S9-perf2-proper-flat-h0/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0658: focused-seeds/geometry/S9-perf2-proper-flat-h0/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume*

(1) *f is proper, flat, and of finite presentation, and*
(2) *for all $s \in S$ we have $\kappa(s) = H^0(X_s, \mathcal{O}_{X_s})$.*

Then we have

(a) *$f_*\mathcal{O}_X = \mathcal{O}_S$ and this holds after any base change,*
(b) *locally on S we have*

$$Rf_*\mathcal{O}_X = \mathcal{O}_S \oplus P$$

in $D(\mathcal{O}_S)$ where P is perfect of tor amplitude in $[1, \infty)$.

Proof. By cohomology and base change (Lemma 0B91) the complex $E = Rf_*\mathcal{O}_X$ is perfect and its formation commutes with arbitrary base change. This first implies that E has tor amplitude in $[0, \infty)$. Second, it implies that for $s \in S$ we have $H^0(E \otimes^{\mathbf{L}} \kappa(s)) = H^0(X_s, \mathcal{O}_{X_s}) = \kappa(s)$. It follows that the map $\mathcal{O}_S \to Rf_*\mathcal{O}_X = E$ induces an isomorphism $\mathcal{O}_S \otimes \kappa(s) \to H^0(E \otimes^{\mathbf{L}} \kappa(s))$. Hence $H^0(E) \otimes \kappa(s) \to H^0(E \otimes^{\mathbf{L}} \kappa(s))$ is surjective and we may apply More on Algebra, Lemma 0A1U to see that, after replacing S by an affine open neighbourhood of s, we have a decomposition $E = H^0(E) \oplus \tau_{\geq 1}E$ with $\tau_{\geq 1}E$ perfect of tor amplitude in $[1, \infty)$. Since E has tor amplitude in $[0, \infty)$ we find that $H^0(E)$ is a flat $\mathcal{O}_S$-module. It follows that $H^0(E)$ is a flat, perfect $\mathcal{O}_S$-module, hence finite locally free, see More on Algebra, Lemma 0658 (and the fact that finite projective modules are finite locally free by Algebra, Lemma 00NX). It follows that the map $\mathcal{O}_S \to H^0(E)$ is an isomorphism as we can check this after tensoring with residue fields (Algebra, Lemma 00O0). □